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A Survey on Handling imbalance Data on Software Defect Prediction

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***Abstract*— Various machine learning techniques for defect software prediction have been used. The learning model is based on the training dataset. The labelled dataset consists of independent variables like difficulty, design complexity, the line of codes, maintenance severity, and dependent variables, either true or false. However, the defect dataset has a problem with class-imbalanced data. Software defects are often rare events compared to non-defective instances. The improvement of the accuracy of defect prediction depends on data quality. Handling imbalanced data is crucial because an imbalanced dataset can lead to biased predictions, where the model fails to accurately predict the minority class, which is the defective software. This study found that most methods used to handle imbalanced data in SDP rely on oversampling, with 13 studies employing this approach. Five studies utilized undersampling, while the remainder combined both techniques or employed alternative methods.**

Keywords—software defect prediction, machine learning, imbalance data.

I. Introduction

Software defect prediction (SDP) is the process to predict the likelihood of defects in different parts of a software system before or during the development process using statistical methods or machine learning. Machine learning is powerful to recognize a case with certain pattern. Software defect prediction benefited from machine learning capability. Supervised machine learning learn the pattern from the training data to build a model. The quality of the model are highly depended on the dataset. In the area of software development, error prevention by predicting software defects in the early stages of development before testing needs to be done to save time and costs [1]

Software defect dataset naturally imbalance, due to the fact that the occurrence of certain cases are dominated in the reality. Imbalance dataset lead to low quality of machine learning model where the model tend to be more capable to detect the majority classes than the minority.

There are some possible solution for imbalance dataset treatment. This paper aims to explore the recent research in imbalance handling on software defect prediction. This paper present a thorough literature reviews. We collect the related paper from 2021- 2024 in recent 4 years.

We found the trend of imbalance dataset handling research are increasing. However, research community still face the challenge on the fact that the study results show that no single method is universally best in all situations; results vary depending on the dataset and approach used. [2] [3]

Based on the imbalance issue mention above, the research try to identify what is the method or technique recently to handling imbalance data in SDP? Does the sampling technique have a significant impact on defect prediction performance? The research structure as following, Background: short description of SDP, SDP process, imbalance data issue, dataset, common problem on SDP, method, related works, what research have been conducted on the imbalance data, Discussion, Conclusion and Reference.

II. Theoritical Background Of Software Defect Prediction

A. Software Defect Prediction Characteristic

The purpose of SDP is to predict whether a software module is defective or defective-free based on several criteria. This technique categorize as binary classification to define which part of module are prone and which are free prone.[4] The process to predict recently using machine learning (ML) algorithm by build defect prediction model. The performance of SDP model depend data quality.

ML require the dataset running in the model distributed equally. In case defect or failure, dataset is imbalanced. Imbalance data make prediction model have trouble to process and the result inaccurate. Developer model have been using numerous strategies to handling imbalance data issue. In order to address the class imbalance problems, data balancing approaches such as oversampling and under sampling are often employed. This study includes SDP papers that have employed strategies to address the problems of data imbalance.

B. Software Defect Prediction Process

The primary stages of SDP can be summed up as follow that involves several steps [5]:

1) gathering both good and bad code samples from software repositories,

2) extracting features to build a dataset,

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3) balancing the dataset if it is unbalanced,
4) training a prediction model on the dataset,
5) predicting the problematic portions of a dataset taken from a new software,
6) assessing the SDP model's performance.

C. Imbalance Class (IC) Chalengge

The reliability and accuracy of predictive models depend on the integrity and completeness of the input data, so data quality is crucial for SDP [6] [7] [8]. One of the main issues with ML and data mining is CI. It describes an imbalance in the percentage of various classes of instances within a dataset. Database rebalancing techniques are frequently employed and there are several approaches to dataset rebalancing.

Under sampling is one technique that balances the dataset by lowering the number of samples from the majority class. Random under sampling is the most basic type of under sampling; it balances the dataset by lowering the samples in the majority class at random. Under sampling has the consequence of potentially erasing crucial information from most class in the training set. Oversampling is an additional technique for dataset rebalancing. A dataset is balanced by oversampling by including a minority class of sample data. Although it randomly duplicates minority class samples, random oversampling (ROS) has the potential to cause significant overfitting.

D. Software Defect Dataset

Table 1 describe available publicly available data set.

TABLE I. A SUMMARY AVAILABLE DATASETS

Dataset	Description		
	Provided By	***No. of Project***	***No. of Metrics***
PROMISE	Open source repository [9]	38	20
NASA	NASA Metrics Data Program (MDP) [10]	13	20-40
SOFTLAB	Software Research Laboratory, Turkey [11]	5	29
Relink	[12]	3	26
AEEEM	[13]	5	61
ECLIPSE1	[14]	3	31
ECLIPSE2	[15]	2	17

III. METHOD

This study conducted to review all research paper about handling imbalance data problem in SDP between 2020 – 2024 using systematic literature review method [16]. The paper collected from science direct, ieeexplore, springer, and IGI Global.

This study based on the following research questions:

- RQ1: what is the method or technique recently to handling imbalance data in SDP.
- RQ2: Does the sampling technique have a significant impact on defect prediction performance.

Since the issue of data imbalance is quite prominent in the SDP area, so it is necessary to review the methods that have been created to overcome data imbalance (RQ1). The RQ2 is posed to determine whether the method used to address CI affect the prediction performance. The process of systematic literature review describes in the figure 1

A. Inclussion Criteria and Exclusion Criteria

The inclusion and exclusion criteria describe in the point below, while exclusion criteria are other than those mentioned in the following point:

- Search process using keywords software defect or fault prediction and imbalance data.
- Written in English
- Full content must be available.
- Published between 2021 –2024
- The results of a search on the google scholar found around 102 open access papers
- Of the total papers, top 31 open access papers were used that were relevant to the given keywords.

The search string used to select the relevant study as follow: *(software OR systems) AND (fault OR defect) AND (prediction OR prone) AND (imbalance data)*.

The step retrieve list of primary study resulted 102 papers from various digital library. Based on the excluding process the final list is 31 papers.

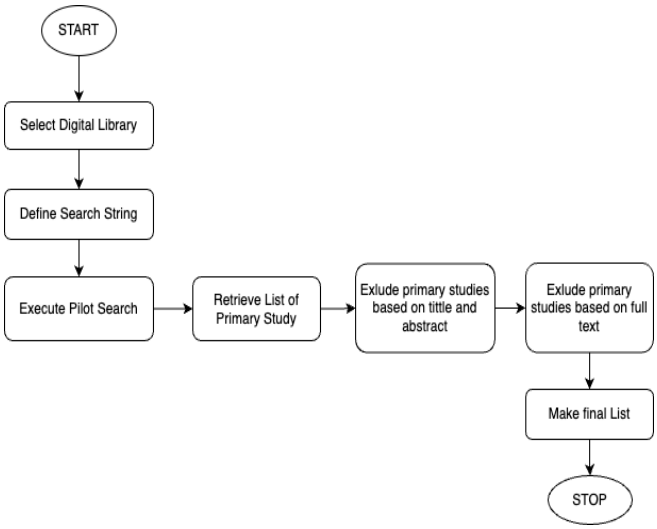


Fig. 1. SLR Process Diagram

IV. RELATED WORK

Pandey and Kumar [4] conducted analysis of the strategies suggested in Software Fault Detection (SFP) from 2019 to 2022. The three SFP techniques covered in this study are ML, DL, and ensemble method based SFP. They conclude that SMOTE is the most popular method of data sampling for addressing problems with data quality.

Liua, Wen, Guangjie, Zhao [17] investigate three types of sampling techniques as resampling, spread subsampling and SMOTE (Synthetic Minority Over-Sampling Technique) and five types of classifier as C4.5, naïve Bayes, logistic regression, support vector machine and DL to study their combined effect prediction. They conclude that there is not

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optimal method among all the 12 combinations in defect prediction.

Pandey and Tripathi [18] address the effects of noise and the CI problem on five baseline SDP models in this work. The different noise levels (0–80%) were manually applied, and their effects on the SDP models' performance were noted. They offer recommendations for the acceptable noise range for baseline models. They proposed the SDP model that has the best noise tolerating ability. The baseline models' True Positive Rate (TPR) and False Positive Rate (FPR) values decrease by 20–30% upon the addition of 10–40% noisy cases. In a similar vein, SDP models' ROC (Receiver Operating Characteristics) values drop to 40–50%. Comparing the proposed model to other conventional model, noise between 40% and 60% is avoided.

Ruchika Malhotra, Shine Kamal [19] study applies sampling techniques and cost-sensitive classifiers to twelve imbalanced NASA datasets to assess the effectiveness of ML classifiers for SDP. They investigate five oversampling techniques, which replicate minority class cases, and they also suggest a novel technique SPIDER3 by proposing adjustments to the SPIDER2 oversampling technique. The outcomes demonstrate how applying oversampling techniques can enhance classifiers and SPIDER3 approach produces positive results. Arun and Lakshmi [20] study the nature of CI problem that exists in the fault prediction data set and to compare the different approaches that have been proposed to address the CI issue. This assessment specifically considers the studies that address problems with data quality.

V. Various Method In Handling Imbalance Data

The popularity of handling imbalance class in SDP are getting increasingly popular among researchers. It is indicated by the positive trend of quantity of papers from 2021 to 2024 which is depicted in figure 2. This study reviews the paper from 2021 to 2024 to focus on the latest innovations in SDP because technology in this field evolves rapidly, and older research may be outdated.

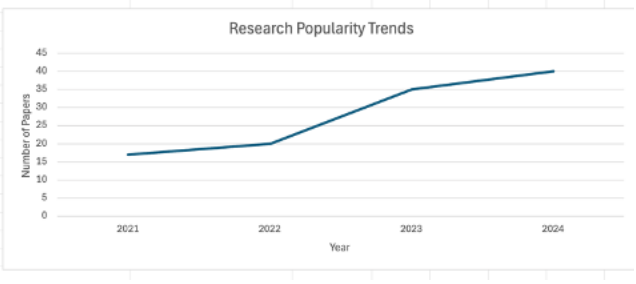


Fig. 2. Research Tren of Data Imbalance Handling

The following study using oversampling (OS) method to handling imbalance data.

Feng S et al. [21] proposed COSTE (Complexity-based Oversampling Technique) to handling imbalance data. This combine the defective instance that have same complexity in generate synthetic instances, makes better in data diversity, prediction performance. Yedida R, Menzies T [22] try to use OS method to datasets prior to implement DL. Usually in deep learning (DL) proses no need first adjusting because leaners engineering feature automatically, make DL work effectively.

Benala et al. focused on enhancing SDP using classifiers on imbalanced data by applying OS methods like ROS, SMOTE, ADASYN, SL-SMOTE, and SVM-SMOTE. The most effective OS method identified was SVM-SMOTE, which increased the prediction performance [23]. Malhotra [24] evaluate OS technique and finds that Adaboost is best, the second is SVM and followed by Bagging Classifier. Singh et al. using Generative OS Method for handling Imbalance data [25].

The other introduces a hybrid model, the SMOTE-SVM (S-SVM) model, which combines Synthetic Minority Oversampling Technique (SMOTE) with Support Vector Machine (SVM) to address the challenge of imbalanced defect datasets. [26] This next paper introduces a RSMOTE-based Data Imbalance Processing (RDIP) model to address the issue of imbalanced data distribution in SDP [27].

Bejjani et al [28] propose CI Reduction (CIR) model to creates symmetry between defect and non-defect records by synthesizing new samples based on centroid calculation of minority class attributes. It outperforms SMOTE in precision, recall, specificity, and geometric mean across various classifiers like AdaBoost, extra tree, K-nearest neighbors, logistic regression, Naive Bayes, and Random Forest. CIR achieve higher accuracy percentages in the range of 82.5% to 92.5% in comparison to SMOTE and K-Means SMOTE.

Wisnu et al. combine SMOTE and neural networks improves accuracy in predictive models. They find that the method involving SMOTE, Artificial Neural Network (ANN), and Association Rule Mining (ARM) enhances predictive accuracy in SDP when contrasted with solely employing the Artificial Neural Network approach [29]. Wisnu et al. (2023) combining the Synthetic Minority Over-sampling Technique (SMOTE) and correlation-based feature selection (CFS) can improve the accuracy. In their experiments using Bayesian Networks (BN) and Decision Tree (DT) classifiers on datasets from NASA, the proposed framework achieved an accuracy of 83.84% and an AUC value of 0.90, indicating its ability to distinguish between defective and non-defective labels without bias. [30]

This paper propose a novel using CNN and GRU with SMOTE Tomek, addressing CI issues in SDP [31]. Results show improvements in prediction accuracy compared to existing, with up to 19% and 24% enhancement in AUC for CNN and GRU models. This study overcome the imbalance issue by balancing the dataset through SMOTE Tomek to reducing overfitting.

Dae-Kyoo Kim et.al [32] using techniques like SMOTE, ADASYN, SMOTE-Tomek, and SMOTE-ENN had a impact on ML models such as DT, RF, LR, and XGB, making them more effective and compatible with ML models compared to DL models. Bashir [33] propose preprocessing method using SMOTE, Fuzzy-rough Instance Selection type II (FRIS-II) and Iterative Noise Filter based on the Fusion of Classifiers (INFFC). The next study present the comparison of OS method such as ROS, ADASYN, SMOTE dan BSMOTE in improving accuracy. The result show that BSMOTE when used with SVM is the best. [34]

The following next study using under sampling or combine both method to handling imbalance data.

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The new algorithm called IDRUS for reducing imbalanced data using under sampling [35]. Akbar et al. using TSC-RUS + B (Two-Step Clustering-Based Under sampling and Bagging Technique). [36]. According to the results, the suggested strategy performs exceptionally well across all datasets (AUC> 0.9). Area under the curve (AUC) was used as the primary evaluation tool for the suggested approach, and five data sets from the NASA metric data program repository were used.

Goyal [7] performed experience on the NASA dataset to find the software defect. An SDP model built using KNN to overcome CI issue. The proposed model call as Neighborhood based Under-Sampling (N-US) with the largest accuracy value of 96.9%, the closest ROC curve to the upper left corner, and the highest AUC score of 95.6%, the suggested model performs better than the other candidate SDP models. At a 95% confidence level, it has the highest statistical prediction power.

The other study propose models that not only predict defects accurately but also offer insights that are easy to understand and apply in practical scenarios, making the models more useful for real-world applications [37]. This study investigate resampling techniques, on interpretation for interpretable models.

Shi et al. [2] proposed US-PONR (Undersampling, Propensity-Score-Matching-Based Oversampling and Noise Reduction) method that reduces CI and noise samples in datasets by OS using propensity score matching (PSM), after eliminating duplicate samples from version iterations through under sampling. When applied to PROMISE improve the datasets quality.Malhotra [38] present experiment several CI handling approach like over dan under sampling dan uses the Random Forest, CNN and LSTM to decide the best. Dataset using PROMISE.

Emy et al. [39] conduct an experiment to optimize SDP in imbalance class by combine resampling method with classification method. The resampling procedure employing OS, US, or hybrid approaches is the first combination of the suggested methods. The second procedure makes use of the logistic regression (LR) and support vector machine (SVM) algorithms for classification. The study demonstrate that substantial results were obtained by combining resampling and the SVM algorithm. The study by Xu et al [40] propose Weighted Extreme Learning Machine (WELM) to address imbalanced data by defining a weight matrix that assigns different weights to majority and minority class samples during classifier training. By adjusting the weights based on the class distribution, WELM can effectively handle imbalanced data.

Golnoush [41] uses the Instance Hardness Threshold (IHT) method for under-sampling, which reduces the majority class data to enhance the model's accuracy. This technique works by identifying and removing instances from the majority class that are considered "hard" or difficult to classify correctly, based on certain criteria or thresholds, showed a 33% improvement in the Area Under the Curve (AUC) metric.

Amanullah et. Al [42] propose method that used to handle the CI problem by generating synthetic samples for the minority class and apply selecting the most relevant features. Finally using Heuristic Learning Vector Quantization (LVQ) Classifier to predict defect with higher accuracy. Other study have explored various techniques and removed multi-collinearity to achieve high accuracy, recall, precision, and ROC AUC scores [43]. Support Vector Classifier, LightGBM, and Logistic Regression, has been essential in optimizing SDP results [44].

Iswafaza et al [45] show using a combination of oversampling (RSMOTE) and undersampling (ENN) methods improves the prediction by addressing data redundancy and CI, resulting in an 11% increase in f-measure evaluation result. The paper [46] introduces three new strategies based on the Optimum-Path Forest (OPF) to handle imbalanced, including OS, US, and a hybrid approach, demonstrating robust performance across various datasets. This next study [47] analysis of various methodologies and classification algorithms. This analysis encompassed algorithms like Random Forest, XGBoost, Support Vector, LightGBM, and Logistic Regression. They also explored over-sampling, under-sampling, and hybridization for achieving dataset balance, in addition to applying pre-processing procedures like mean value imputation and min-max scaling on the AEEEM dataset. Ponnala et al 2023 [48] using various methods to predict defects, apply different techniques also removing multi-collinearity from the data then achieved impressive performance metrics, including 92% accuracy, 96% recall, 89% precision, and a 0.92 ROC AUC score.

VI. Disscusion

The research discusses several studies that suggest different approaches to tackle the problem of CI in SDP. An approach by Shi et al. [2] is the US-PONR method, which integrates US with propensity score matching to enhance dataset quality through reducing imbalance and noise. Akbar et al. [36] applied TSC-RUS + B, demonstrating strong performance with an AUC exceeding 0.9 on five NASA datasets. Goyal presented the Neighborhood-based Under-Sampling (N-US) technique, achieving an impressive accuracy of 96.9% and an AUC of 95.6%. Emy et al. combined resampling techniques with classification algorithms like LR and SVM, yielding significant outcomes.

The COSTE method was formulated by Feng S et al.[21], merging intricate instances to produce synthetic ones. Yedida [22] and Menzies employed OS prior to implementing DL, showcasing effectiveness. Bejjani et al. [28]introduced the CIR model, which surpassed SMOTE across various metrics. Wisnu et al. discovered that the combination of SMOTE with ANN and ARM enhances predictive accuracy. These studies highlight that a variety of resampling techniques can boost the SDP by addressing CI.

The paper limitation is primarily investigating certain OS and US techniques, which may not encompass all possible methods available for addressing class imbalance in software defect prediction. Table II show the summary of the state-of-the-art research in imbalance data handling on SDP.

VII. Conclusion

The discussion of these methods highlights the advancements in handling imbalanced data, in the context of SDP. Each method offers unique benefits and demonstrates improvements in various performance metrics, such as accuracy, AUC, precision, and recall as described on table II. The combination of resampling techniques with advanced ML

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and DL shows promising results in improve defect prediction. Future research could focus on further refining these methods, exploring new hybrid approaches, and evaluating their performance across diverse datasets.

TABLE II. IMBALANCE HANDLING METHOD IN VARIOUS RESEARCH

Paper	Dataset	Imbalance Handling	Machine Learning Method	Achievement
[2]	PROMISE	US-PONR *	SVM, RF	Improve quality dataset 10%
[36]	NASA	US & bagging *	NB, KNN, SVM, LR	Improve AUC > 0.9
[7]	NASA	N-US *	ANN, SVM, DT, KNN, NB,	AUC= 95,6%, Accuracy = 96%
[39]	NASA	US, OV, Hybrid *#+	SVM,LR	SV Improve AUC= 17,19%; LR improve AUC=7%
[21]	PROMISE	COSTE #	SVM, KNN, RF, MLP	COSTE better than SMOTE
[22]	PROMSIE	Fuzzy Sampling	Deep Belief Network	Improve accuracy
[38]	PROMISE	OV and US *#	RF, SVM, LSTM	Improve AUC
[25]	PROMISE, JIRA, Eclipse	generative oversampling #	generative adversarial network	better than other imbalance method.
[28]	PROMISE	Class Imbalance Reduction (CIR)	KNN. NB. LR, RF	Improve SMOTE and K-Means SMOTE
[29]	PROMISE	SMOTE	ANN and ARN	accuracy increase are 2.2%, 18.4%, 22.4% and 8.8%
[30]	NASA	SMOTE	Bayesian Network, Decision Tree (DT)	Improve accuracy
[23]	NASA	SMOTE, ADASYN, SVM-SMOTE, Safe-level SMOTE	DT, NB, RF, Adaboost, Bagging	Accuracy 80%
[31]	PROMISE	SMOTE Tomek	CNN, Gated Recurrent Unit	Increase 19% for the CNN and 24% for the GRU
[40]	NASA	KPWE	RF,NB,SVM,LR	KPWE superior in F-measure, AUC, MCC
[26]	PROMISE, AEEEM	SMOTE-SVM	SVM	The proposed S-SVM model outperforms other models
[27]	NASA	RSMOTE	SVM, ANN	6.98% improve
[35]	PROMISE	IDRUS	various classification techniques applied	improves performance of ML
[49]	PROMISE	Structured Under-sampling	C4.5,LR, NB, SVM, RF, Ripper, IBk	Better performance than Random Under Sampling
[37]	47 public dataset	SMOTE	C4.5, Ripper	rule-based interpretable models able to deal with imbalanced data
[41]	NASA	Instance Hardness Threshold	SVM	33% improvement in AUC metric
[42]	PROMISE	generating synthetic samples	Heuristic Learning Vector Quantization (LVQ)	Improve accuracy
[32]	NASA	SMOTE, ADASYN, SMOTE-Tomek, and SMOTE-ENN	DT, RF, LR, XGB, MLP	balancing positive on MLP, negative on CNN and LSTM, positive in all ML
[43]	NASA	Hellinger distance (HD	HDDT and ANN	improve data quality and SDP accuracy
[44]	9 real-projects.	CFIWSE	Specific classifiers are not detailed	Improve accuracy
[45]	NASA	COU model using RSMOTE , ENN	Deep Learning	Enhancing performance 11%.
[46]	UCI	Optimum-Path Forest (OPF)	CNN	Improve F1 score
[33]	AEEEM	FRIS-II and INFFC	C4.5 NB RF	Improve AUC
[47]	AEEEM	class-weight-based, OV, US	RF, XGBoost, SVM, LightGBM, LR	Improve F1 score, accuracy, AUC
[48]	NASA	removed multi-collinearity	Specific classifiers are not detailed	92% Accuracy, 96% Recall, 89% Precision, and a 0.92 ROC AUC Score
[34]	NASA	ROS, ADASYN, SMOTE, and B-SMOTE	NB, SVM	Improve AUC and ROC

REFERENCES

[1] E. K. Ali, M. M. Eissa, and A. F. Omara, "A Review for Software Defect Prediction Using Machine Learning Algorithms," in *Green Sustainability: Towards Innovative Digital Transformation*, D. Magdi, A. A. El-Fetouh, M. Mamdouh, and A. Joshi, Eds., Singapore: Springer Nature Singapore, 2023, pp. 219–231.

[2] H. Shi, J. Ai, J. Liu, and J. Xu, "Improving Software Defect Prediction in Noisy Imbalanced Datasets," *Applied Sciences (Switzerland)*, vol. 13, no. 18, Sep. 2023, doi: 10.3390/app131810466.

[3] R. Malhotra and S. Kamal, "An empirical study to investigate oversampling methods for improving software defect prediction using imbalanced data," *Neurocomputing*, vol. 343, pp. 120–140, May 2019, doi: 10.1016/j.neucom.2018.04.090.

[4] S. Pandey and K. Kumar, "Software Fault Prediction for Imbalanced Data: A Survey on Recent Developments," in *Procedia Computer Science*, Elsevier B.V., 2022, pp. 1815–1824. doi: 10.1016/j.procs.2023.01.159.

[5] G. Giray, K. E. Bennin, Ö. Köksal, Ö. Babur, and B. Tekinerdogan, "On the use of deep learning in software defect prediction," *Journal of Systems and Software*, vol. 195, Jan. 2023, doi: 10.1016/j.jss.2022.111537.

[6] Kirti Bhandari, "Data quality issues in software fault prediction: a systematic literature review," *Artificial Intelligence* , pp. 7839–7908, 2023.

[7] S. Goyal, "Handling Class-Imbalance with KNN (Neighbourhood) Under-Sampling for Software Defect Prediction," *Artif Intell Rev*, vol. 55, no. 3, pp. 2023–2064, Mar. 2022, doi: 10.1007/s10462-021-10044-w.

[8] G. Haixiang, L. Yijing, J. Shang, G. Mingyun, H. Yuanyue, and G. Bing, "Learning from class-imbalanced data: Review of methods and applications," May 01, 2017, *Elsevier Ltd*. doi: 10.1016/j.eswa.2016.12.035.

[9] "TERA PROMISE HOME," https://openscience.us/repo/.

[10] J. Pachouly, S. Ahirrao, K. Kotecha, G. Selvachandran, and A. Abraham, "A systematic literature review on software defect prediction using artificial intelligence: Datasets, Data Validation Methods, Approaches, and Tools," May 01, 2022, *Elsevier Ltd*. doi: 10.1016/j.engappai.2022.104773.

[11] B. Turhan, T. Menzies, A. B. Bener, and J. Di Stefano, "On the relative value of cross-company and within-company data for defect prediction," *Empir Softw Eng*, vol. 14, no. 5, pp. 540–578, Oct. 2009, doi: 10.1007/s10664-008-9103-7.

[12] R. Wu, H. Zhang, S. Kim, and S.-C. Cheung, "ReLink: recovering links between bugs and changes," in *Proceedings of the 19th ACM SIGSOFT Symposium and the 13th European Conference on Foundations of Software Engineering*, in ESEC/FSE '11. New York, NY, USA: Association for Computing Machinery, 2011, pp. 15–25. doi: 10.1145/2025113.2025120.

[13] M. D'Ambros, M. Lanza, and R. Robbes, "Evaluating defect prediction approaches: a benchmark and an extensive comparison," *Empir Softw Eng*, vol. 17, no. 4, pp. 531–577, 2012, doi: 10.1007/s10664-011-9173-9.

[14] T. Zimmermann, R. Premraj, and A. Zeller, "Predicting Defects for Eclipse," in *Proceedings of the Third International Workshop on Predictor Models in Software Engineering*, in PROMISE '07. USA: IEEE Computer Society, 2007, p. 9. doi: 10.1109/PROMISE.2007.10.

[15] S. Kim, H. Zhang, R. Wu, and L. Gong, "Dealing with noise in defect prediction," in *2011 33rd International Conference on Software Engineering (ICSE)*, 2011, pp. 481–490. doi: 10.1145/1985793.1985859.

[16] H. Zhang *et al.*, *EAST'12 : proceedings of the 2nd International Workshop on Evidential Assessment of Software Technologies : September 22, 2012, Lund, Sweden*.

[17] Y. Liu, W. Zhang, G. Qin, and J. Zhao, "A comparative study on the effect of data imbalance on software defect prediction," in *Procedia Computer Science*, Elsevier B.V., 2022, pp. 1603–1616. doi: 10.1016/j.procs.2022.11.349.

[18] S. P. Sushant Pandey, "An empirical study toward dealing with noise and class imbalance issues in software defect prediction," *Data Analytic and Machine Learning*, vol. 25, pp. 13465–13492, 2021.

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[19] R. Malhotra and S. Kamal, "An empirical study to investigate oversampling methods for improving software defect prediction using imbalanced data," *Neurocomputing*, vol. 343, pp. 120–140, May 2019, doi: 10.1016/j.neucom.2018.04.090.

[20] C. Arun and C. Lakshmi, "Class Imbalance in Software Fault Prediction Data Set," *Artificial Intelligence and Evolutionary Computations in Engineering Systems*, vol. 1056, pp. 745–756, 2020.

[21] S. Feng *et al.*, "COSTE: Complexity-based OverSampling TEchnique to alleviate the class imbalance problem in software defect prediction," *Inf Softw Technol*, vol. 129, Jan. 2021, doi: 10.1016/j.infsof.2020.106432.

[22] R. Yedida and T. Menzies, "On the Value of Oversampling for Deep Learning in Software Defect Prediction," Aug. 2020, [Online]. Available: http://arxiv.org/abs/2008.03835

[23] T. R. Benala and K. Tantati, "Efficiency of oversampling methods for enhancing software defect prediction by using imbalanced data," *Innov Syst Softw Eng*, vol. 19, no. 3, pp. 247–263, Sep. 2023, doi: 10.1007/s11334-022-00457-3.

[24] R. Malhotra, V. Kashyap, and V. Sharma, "Comparative study of Sampling Techniques for Software Defect Prediction," in *8th International Conference on Advanced Computing and Communication Systems, ICACCS 2022*, Institute of Electrical and Electronics Engineers Inc., 2022, pp. 760–766. doi: 10.1109/ICACCS54159.2022.9785238.

[25] Cite This PDF Santosh Singh Rathore; Satyendra Singh Chouhan; Dixit Kumar Jain; Aakash Gopal Vachhani, "Generative Oversampling Methods for Handling Imbalanced Data in Software Fault Prediction," *IEEE*, 2022.

[26] M. Mustaqeem and T. Siddiqui, "A hybrid software defects prediction model for imbalance datasets using machine learning techniques: (S-SVM model)," *Journal of Autonomous Intelligence*, vol. 6, no. 1, pp. 1–19, 2023, doi: 10.32629/jai.v6i1.559.

[27] Cite This PDF Yang Qu; Zhenming Li; Jiaoru Zhao; Hui Li, "Unbalanced data processing for software defect prediction," IEEE, 2022.

[28] K. K. Bejjanki, J. Gyani, and N. Gugulothu, "Class imbalance reduction (CIR): A novel approach to software defect prediction in the presence of class imbalance," *Symmetry (Basel)*, vol. 12, no. 3, Mar. 2020, doi: 10.3390/sym12030407.

[29] Cite This PDF Wisnu Arya Dipa; Wikan Danar Sunindyo, "Software Defect Prediction Using SMOTE and Artificial Neural Network," *IEEE*, 2021.

[30] J. H. Abawajy, K.-K. Raymond, C. Haruna, and C. Editors, "Lecture Notes in Networks and Systems 254 International Conference on Emerging Applications and Technologies for Industry 4.0 (EATI'2020) Emerging Applications and Technologies for Industry 4.0." [Online]. Available: http://www.springer.com/series/15179

[31] N. A. A. Khleel and K. Nehéz, "A novel approach for software defect prediction using CNN and GRU based on SMOTE Tomek method," *J Intell Inf Syst*, vol. 60, no. 3, pp. 673–707, Jun. 2023, doi: 10.1007/s10844-023-00793-1.

[32] D.-K. Kim and Y. K. Chung, "Addressing Class Imbalances in Software Defect Detection," *Journal of Computer Information Systems*, vol. 64, no. 2, pp. 219–231, 2024, doi: 10.1080/08874417.2023.2187483.

[33] K. Bashir, T. Li, C. W. Yohannese, and M. Yahaya, "SMOTEFRIS-INFFC: Handling the challenge of borderline and noisy examples in imbalanced learning for software defect prediction," *Journal of Intelligent and Fuzzy Systems*, vol. 38, no. 1, pp. 917–933, 2020, doi: 10.3233/JIFS-179459.

[34] S. N. and Y. Y. and I. F. and P. A. A. and S. E. N. and H. T. Riston Theodorus and Suherman, "Oversampling Methods for Handling Imbalance Data in Binary Classification," in *Computational Science and Its Applications – ICCSA 2023 Workshops*, B. and R. A. M. A. C. and G. C. and S. F. and K. Y. and T. C. M. Gervasi Osvaldo and Murgante, Ed., Cham: Springer Nature Switzerland, 2023, pp. 3–23.

[35] S. Manjula B. and Layaq, "A New Method for Imbalanced Data Reduction Using Data Based Under Sampling," in *Soft Computing and Signal Processing*, V. K. and W. J. and R. K. T. V Reddy V. Sivakumar and Prasad, Ed., Singapore: Springer Nature Singapore, 2023, pp. 243–250.

[36] M. Faittullah Akbar, I. Kurniawan, and A. Fauzi, "Mengatasi Imbalanced Class Pada Software Defect Prediction Menggunakan Two-Step Clustering-Based Undersampling dan Bagging Tehcnique," *JURNAL INFORMATIKA*, vol. 6, no. 1, pp. 107–113, 2019, [Online]. Available: http://ejournal.bsi.ac.id/ejurnal/index.php/ji/article/view/5448

[37] Y. Gao, Y. Zhu, and Y. Zhao, "Dealing with imbalanced data for interpretable defect prediction," *Inf Softw Technol*, vol. 151, Nov. 2022, doi: 10.1016/j.infsof.2022.107016.

[38] S. J. & U. J. Ruchika Malhotra, "Using Deep Learning and Class Imbalance Techniques to Predict Software Defects," International Conference on IoT Based Control Networks and Intelligent Systems, 2024.

[39] by S. Emy, "Optimization of software defects prediction in imbalanced class using a combination of resampling methods with support vector machine and logistic regression," 2023.

[40] Z. Xu *et al.*, "Software defect prediction based on kernel PCA and weighted extreme learning machine," *Inf Softw Technol*, vol. 106, pp. 182–200, Feb. 2019, doi: 10.1016/j.infsof.2018.10.004.

[41] W. Z. T. J. Z. W. T. E. S. J. H. Golnoush Abaei, "Improving software fault prediction in imbalanced datasets using the under-sampling approach," ICSCA , 2022.

[42] M. Amanullah, S. Thanga Ramya, M. Sudha, V. P. Gladis Pushparathi, A. Haldorai, and B. Pant, "Data sampling approach using heuristic Learning Vector Quantization (LVQ) classifier for software defect prediction," *Journal of Intelligent & Fuzzy Systems*, vol. 44, pp. 3867–3876, 2023, doi: 10.3233/JIFS-220480.

[43] A. K. Chakraborty and B. Karmakar, "Software Defect Prediction Through a Hybrid Approach Comprising of a Statistical Tool and a Machine Learning Model," in *Applications of Operational Research in Business and Industries*, A. Gunasekaran, J. K. Sharma, and S. Kar, Eds., Singapore: Springer Nature Singapore, 2023, pp. 1–19.

[44] J. Xu, J. Shang, and Z. Huang, "CFIWSE: A Hybrid Preprocessing Approach for Defect Prediction on Imbalance Real-World Datasets," in *2022 IEEE 22nd International Conference on Software Quality, Reliability, and Security Companion (QRS-C)*, 2022, pp. 392–401. doi: 10.1109/QRS-C57518.2022.00064.

[45] A. F. Iswafaza and S. Rochimah, "Software Defect Prediction Using a Combination of Oversampling and Undersampling Methods," in *2022 6th International Conference on Information Technology, Information Systems and Electrical Engineering (ICITISEE)*, 2022, pp. 127–132. doi: 10.1109/ICITISEE57756.2022.10057798.

[46] L. A. Passos, D. S. Jodas, L. C. F. Ribeiro, M. Akio, A. N. de Souza, and J. P. Papa, "Handling imbalanced datasets through Optimum-Path Forest," *Knowl Based Syst*, vol. 242, p. 108445, 2022, doi: https://doi.org/10.1016/j.knosys.2022.108445.

[47] R. Bafna, R. Jain, and R. Malhotra, "A Comparative Study of Classiication Techniques and Imbalanced Data Treatment for Prediction of Software Faults A Comparative Study of Classification Techniques and Imbalanced Data Treatment for Prediction of Software Faults," 2023, doi: 10.21203/rs.3.rs-2809140/v1.

[48] R. Ponnala and C. R. K. Reddy, "Hybrid Model to Address Class Imbalance Problems in Software Defect Prediction using Advanced Computing Technique," in *2023 2nd International Conference on Applied Artificial Intelligence and Computing (ICAAIC)*, 2023, pp. 1115–1122. doi: 10.1109/ICAAIC56838.2023.10141379.

[49] Cite This PDF Syed Fawad-ul-Hassan Gillani; Aamer Nadeem; Muhammad Rizwan, "An Effective Undersampling Approach to Deal with Class Imbalance Problem in Software Defect Prediction," 2022.

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